# AgentWorkShop: A Reference Implementation for Task-to-Device Integration in Industrial Agents

***Abstract*—Industrial agents need a software path that connects a task to production context, semantic operations, and configured device interfaces. AgentWorkShop is a modular reference implementation that composes task services, registered harnesses, line–product–recipe–run context, per-call operation checks, semantic I/O, protocol drivers, and service-owned records. It resolves context for each agent turn and checks applicable authority at dispatch; prompt context grounds a proposal but is not a task-pinned authorization token. We evaluated one registered harness in cast-film and annealing simulators with different signal and protocol mappings. Six sequential tasks received synthetic approval and readback-verified receipts. After each receipt, a 20-s delay preceded an explicit probe that was ingested and queried through an isolated SQLite time-series fallback; task state was checked after the query. Each probe was linked to its task and approval records and tagged with its active production run. All six samples remained outside their target bands. A separate deterministic fixture exercised one task per scene without an external language model; all 31 runner assertions passed. These results establish the tested software paths, not process attainment, normal online DAQ operation, cross-harness portability, or comparative benefit.**

***Index Terms*—Industrial agents, industrial communication, industrial informatics, production systems, semantic data integration.**

## I. Introduction

INDUSTRIAL-AGENT deployments must connect each task to its production context, process signals, and equipment interfaces. Existing work addresses end-to-end automation control, operator-reviewed control-sequence generation, modular factory skills, and production scheduling [1]–[4]. Other studies target PLC code generation, context-aware control, resource capability discovery, shop-floor coordination, digital-twin fault handling, natural-language machine access, and robotic-line control [5]–[10]. Across production lines, agent roles, signal maps, protocols, objectives, and product/run context vary. An integration architecture must accommodate these differences around shared task and operation services while keeping each action tied to its active production context. Recent industrial-agent architectures add AAS-grounded OPC UA actions with signed approval, P&ID-grounded proposal checks, and governed sensor–MES–ERP evidence streams [11]–[13]. These works address asset-specific execution, process validation, or closed-loop analytics. This paper addresses the complementary software-integration question of composing task and harness routes, production context, semantic protocol I/O, and route-owned records. Its simulator evaluation tests configured paths rather than process-control performance or portability.

Consider two production scenes with different process signals and interfaces: cast-film thickness control and annealing-zone temperature control. An agent task in either scene needs production context, typed signal access, and a route from a named operation to the configured device protocol. The framework must also distinguish a command receipt from a subsequent process observation. The shared task and operation services can therefore be configured with scene-specific signal maps, protocols, and limits while retaining a common execution path. The evaluation instantiates this path in two simulator scenes.

This paper presents AgentWorkShop as a reference implementation of task-to-device integration for industrial agents. It connects collaborative tasks and registered harnesses to production context, manufacturing execution system (MES) data, analysis and modeling services, data-acquisition (DAQ) and write-control (DCW) nodes, and protocol-specific drivers. Its interfaces include command-line, web, SDK, REST, Model Context Protocol (MCP), and Agent-to-Agent (A2A) surfaces. Task and chat routes enter the collaboration runtime, whereas a direct REST tool-invocation route can call the shared operation dispatcher without creating a task. This direct entry path is described from the implementation but was not exercised in R20 or R21 (Table II). Scene blueprints define signal maps, objectives, and guards; channel profiles, tool policies, and agent bindings scope operations. Agents and callers invoke named operations through shared runtime checks; the framework does not choose routes autonomously and retains proposals, command confirmation, and process observations as distinct records. Fig. 1 presents the system architecture, and Fig. 2 shows the cross-module task flow. Section IV details the node-exploration and recipe lifecycles.

This paper asks whether a configured registered-harness path can execute bounded tasks in two simulator scenes with different signal and protocol mappings while retaining a checkable relation from task through approval and verified route receipt to a subsequent run-tagged measurement. A deterministic fixture separately replays the software route without external model decisions. These evaluations test the configured paths and their records; they do not test configuration reuse, cross-harness portability, or an immutable context snapshot across prompt, approval, and dispatch. They assess path feasibility and record linkage, not benefit over alternative integration designs or configuration-change effort.

The paper makes two contributions:

1) It specifies and implements a modular task-to-device reference path that joins collaborative tasks and registered harnesses to turn-time production context, per-call operation checks, semantic I/O, configured protocol drivers, and service-owned records. Its design distinction is the explicit separation of grounding context, dispatch authority, device receipt, and later process observation. This is a software-architecture contribution; it introduces

no new protocol, safety-verification method, or process-control algorithm.

2) It evaluates selected paths in cast-film and annealing simulator scenes. In R20, six sequential tasks through one registered harness passed the task-to-approval-to-receipt-to-sample linkage checks; none of the six samples reached its target band. In R21, a deterministic fixture executed one task per scene with no external model and passed 31 runner assertions, including cleanup checks. These results verify configured software paths, not process attainment, cross-harness portability, or performance superiority.

Section II reviews related industrial-agent and integration work. Section III presents the design rationale, architecture, and cross-module execution flow; Section IV explains module responsibilities and their integrated operation. Section V describes the experiments and results, and Sections VI and VII discuss the evidence limits and conclude the paper.

## II. Related Work

This section relates AgentWorkShop to plant-facing industrial agents and to production and automation abstractions for context, tool access, and operational constraints. Table I compares integration and execution features described in selected industrial-agent architectures, including systems with adjacent but distinct process-control objectives. The comparison is focused rather than exhaustive; a feature marked as not specified was not described in the cited study and is not inferred to be absent. Studies were included in this comparison when they describe an industrial agent's production context, plant-facing operations, or an action, approval, or evidence pathway. This criterion defines the table's scope; it is not a comprehensive survey or a performance ranking.

### A. Industrial Agents and Plant Integration

Industrial-agent systems, including large language model (LLM)-based systems, span end-to-end automation control, supervisory services, programmable logic controller (PLC) interfaces, manufacturing coordination, and operator-facing machine access. Xia et al. present an LLM agent framework for end-to-end industrial automation control, using structured prompts and event-driven context for production planning and operation [1]. Huang et al. present a multimodal architecture whose stated focus is automated control generation for industrial cyber-physical systems [2]. Biskupovic et al. study agent-based supervision for service-oriented industrial systems [14]; Agents4PLC and InstructMPC address PLC code generation and human–LLM context-aware control, respectively [5], [6]. Manufacturing studies consider dynamic resource-capability exploration [7], multi-agent shop-floor coordination [8], and digital-twin-supported fault handling [9]. Natural-language machine access [10] and real-time LLM control of a robot-operated serial line [4] extend this work to operator interfaces and robotic-line control. Together, these studies address different operational contexts, interfaces, and allocations of responsibility between agents, operators, and controllers.

The closest architecture-level comparison is Xia et al., who connect cognitive, digital-twin, and physical-automation layers and organize twin information as state snapshots, event messages, and plan sequences; their case studies address robotic automation and process simulation [15]. Other nearby designs expose factory-module operations through MCP with state-based tool filtering, plan constrained manufacturing steps, or compose semantic recipes from OPC UA skills [3], [16], [17]. These studies describe important parts of plant-facing integration. The recent systems in Table I also combine agent execution with approval, industrial data, or audit records. Graul et al. ground proposals in the Asset Administration Shell and route approved actions through OPC UA Methods, with signed approvals and audit logging [11]. Schall separates advisory language-model agents from P&ID-grounded verification and execution, and evaluates structural checks across five simulated process scenarios [12]. Lin and Chen build a governed sensor–MES–ERP digital thread for closed-loop monitoring and control, evaluated on real manufacturing records [13]. These works address asset-specific approval, process validation, or closed-loop analytics. AgentWorkShop's narrower architectural delta is the composition of task/chat execution and direct operation calls with registered harnesses, production context, semantic drivers, shared per-call dispatch, and service-owned operation records. The comparison does not imply that other systems lack these capabilities. R20 and R21 test selected routes through this composition; neither tests comparative utility or generality beyond the configured paths.

The evaluation unit is an execution record whose required identifiers link task, approval, active run, verified receipt, and subsequent sample; other service records may expose only the identifiers appropriate to their route. This record path is a concrete implementation focus, not a claim that prior systems cannot provide equivalent linkage. AgentWorkShop adds no new process control or safety-verification method. Authority checks and command confirmation complement safety-aware decision support [18]; they do not replace process protection or establish operator effectiveness.

### B. Production Context, Agent Interfaces, and Automation

Industrial integration also depends on abstractions that are not specific to agent architectures. MCP standardizes tool discovery and calls with request-scoped authorization [19], whereas ISA-88 and ISA-95 describe recipes and production operations [20], [21]. Run-to-run methods connect process observations to successive production runs [22]. At the agent level, AgentSpec and CaMeL address tool behavior and capability-based information flow [23], [24]. Together, these works provide complementary abstractions for production context, tool interaction, and information flow. AgentWorkShop composes such abstractions with its task runtime, operation checks, semantic I/O, and route-specific records; this paper evaluates selected paths through that composition rather than proposing a replacement for the underlying standards.

Industrial security and record requirements further constrain that architecture: FDA guidance discusses trustworthy

TABLE I
FEATURE-LEVEL COMPARISON OF SELECTED PLANT-FACING INTEGRATION STUDIES. "NOT SPECIFIED" MEANS THAT THE CITED STUDY DOES NOT DESCRIBE THE FEATURE; IT IS NOT EVIDENCE OF ABSENCE. THE TABLE COMPARES STATED DESIGNS AND EVALUATION EVIDENCE, NOT PERFORMANCE.

| System | Task and production context | Per-call operation checks | Device-facing route | Reported evaluation and record link |
|---|---|---|---|---|
| Xia et al. [15] | Cognitive, twin, and physical layers; snapshots, events, and plans | Not specified | Digital-twin/physical-automation integration | Robotic automation and process-simulation case studies |
| Köhle et al. [3] | Factory-module agents with injected live state | State-based tool filtering | Per-module MCP exposes OPC UA methods; MQTT coordination | Nine simulated production challenges |
| Vieira da Silva et al. [16] | LLM plans manufacturing steps under constraints | Not specified | MCP-exposed manufacturing resource functions | Laboratory multi-step process with constraint handling |
| Köhle et al. [17] | AI-assisted recipes composed from OPC UA skills | Not specified | OPC UA skills and virtual commissioning | Unity assembly-workstation proof of concept |
| Graul et al. [11] | AAS-grounded asset and production context | Signed HITL approval; immutable audit log | PLC-validated OPC UA Methods | Two laboratory PoCs; single scripted runs |
| Schall [12] | P&ID, SCADA, time-series, and SOP context | Deterministic topology and rule checks | Advisory proposal, gated execution | Five simulated scenarios; 43 injected proposals and 50 LLM runs |
| Lin and Chen [13] | Sensor–MES–ERP digital thread | SOP-, risk-, and capability-aware gating | Edge monitoring and closed-loop actions | 25,275 real records; monitoring and operational metrics |
| This work | Collaborative tasks and direct calls; per-turn line–product–recipe–run context | Scope, policy, grant/binding at dispatch; route-specific state rechecked before write | Semantic node resolves to configured DAQ/DCW driver | R20: 6/6 live-agent traces; R21: 2/2 deterministic fixture traces across two mappings; no process attainment claim |

records and audit trails [25], IEC 62443-3-3 specifies system-security requirements [26], and DNP3 includes select-before-operate [27]. Control-assurance methods address state evolution through shielding, control-barrier functions, or Simplex architectures [28]–[30]. These standards and methods inform the boundaries within which an industrial-agent framework must operate; they are not contributions claimed by AgentWorkShop.

The evaluation tests selected software paths and simulator scenes, not physical-plant performance.

## III. SYSTEM DESIGN

AgentWorkShop is designed for industrial tasks whose agents, production contexts, interfaces, and process signals vary between deployments. Its architecture organizes these variations around shared task and operation services, checks each proposed operation against the active production context before device routing, and retains route-specific evidence that distinguishes a command receipt from a measured process response.

### A. Design Rationale and Requirements

Five requirements—scene variability, harness variability, operation authority, outcome observability, and extensibility—organize six design rules (D1–D6). These are deployment constraints drawn from the target workflows. D1 stores scene settings as versioned data; D2 provides registered harness, tool, and driver interfaces; D3 separates proposal from execution; D4 treats the tool catalog as discovery rather than permission; D5 routes actuator writes through the common DCW admission point; and D6 retains evidence by operation.

The requirements trace to implementation as follows. Scene variability maps to D1, scene blueprints, and context resolution; harness variability and extensibility map to D2, the registered harness and driver interfaces; operation authority maps to D3–D5, the shared dispatcher, grants, bindings, and write admission; outcome observability maps to D6, route-specific receipts, and observations. R13 exercises two scene-specific protocol and data paths; R20 exercises the configured task-to-device path through one registered harness, synthetic approvals, and route-specific records. Section V reports the experimental scope and evidence limits. AgentWorkShop supplies task and operation services; agent strategy and independent process protection remain outside this design.

### B. Layered Architecture

Fig. 1 maps the reference implementation onto six bands and their route-specific evidence records. The access band exposes command-line, web, SDK, REST, MCP, and A2A entry surfaces. Task and chat routes enter the collaboration band, which contains workspaces, channels, task queues, shared memory, mailboxes, task templates, and harness adapters. A direct REST tool-invocation route can call the shared operation dispatcher without creating a task or invoking a harness; MCP

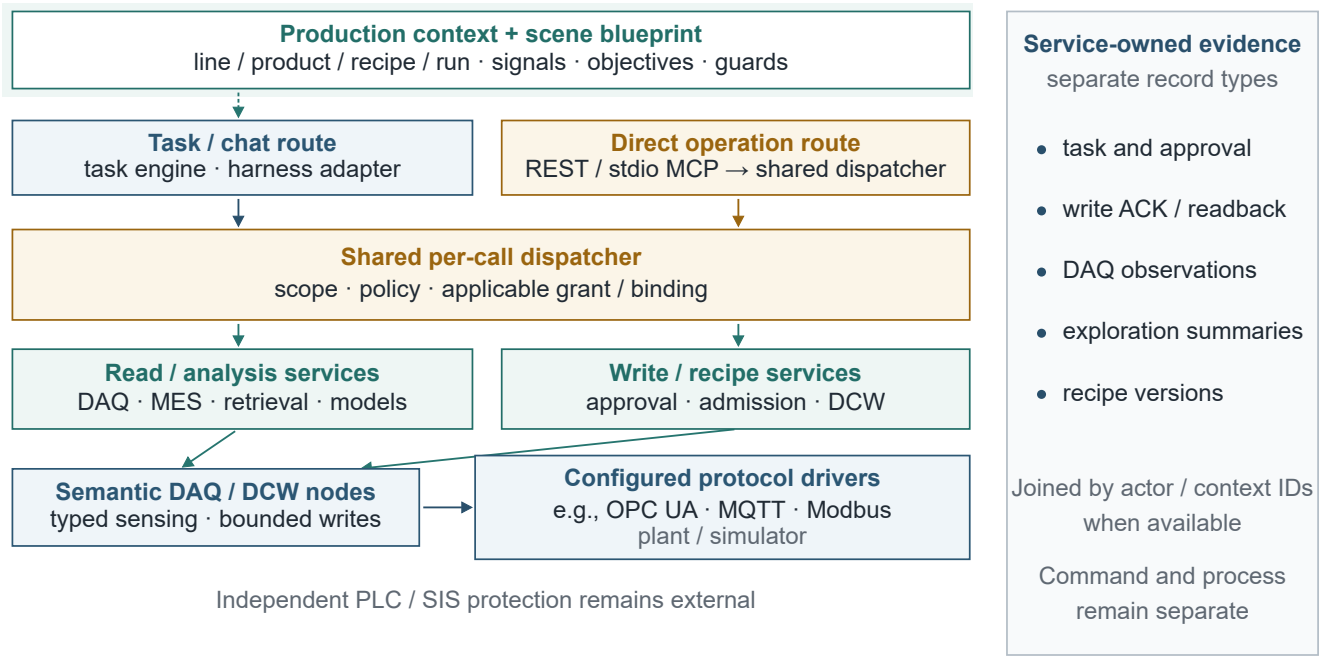



Fig. 1. AgentWorkShop architecture and record ownership. Task/chat calls pass through collaboration and a registered harness, whereas direct REST/stdio MCP calls enter the shared dispatcher; the direct entry is implemented but was not exercised in this study. Scene configuration and production context inform task execution; operation handlers apply current route checks. Semantic I/O resolves through configured drivers, while each service retains its own records. Independent PLC/SIS protection remains external.

also includes separate management and task operations. The governance band projects an operation catalog and checks each invocation. The context band assembles bound-line and active-run information for prompt grounding, while operation handlers resolve the live state relevant to their route. The service band provides acquisition, write control, analysis, retrieval, and optional model services. The field-I/O band resolves semantic nodes to built-in or plugin drivers for a plant or simulator.

Each band has a distinct responsibility. Task/chat routes enter the task runtime; direct operation routes enter the dispatcher; and both use shared operation checks where applicable. A harness adapter supplies agent-specific interaction on task paths, semantic nodes hide protocol details from operations, and the driver registry provides the device-facing extension point. Owning services retain task, command, observation, and version records separately and link them to available actor and production context. No single status conflates these outcomes.

### C. Cross-Module Execution Flow

Fig. 2 summarizes a task turn from entry through operation execution, with route-specific evidence retained by owning services. Registered harness adapters receive a scoped tool catalog during setup and supported registry refreshes; the runtime assembles prompt-time production context for each agent turn. This context is grounding information, not a task-pinned transaction. For each returned tool call, the shared dispatcher rechecks invocation scope and channel policy and, for industrial operations, current grant and binding conditions before handler selection. The selected handler then applies route-specific guards, active-run checks, approval revalidation, and device-I/O steps as applicable. Command receipts and later process observations remain distinct records.

For an industrial operation $u$ by agent $a$ in channel $c$, define the shared dispatch condition as

$$\mathcal{A}(a, c, u) = S(a, c, u) \wedge P(c, u) \wedge G(a, u) \wedge B(a, u), \quad (1)$$

where $S$, $P$, $G$, and $B$ denote invocation scope, current channel policy, runtime grant, and agent binding/resource ownership. Scope and channel policy are checked for every host-tool call. The grant check applies to industrial tools except the explicit grant-exempt set; the binding/resource check applies when the operation declares resource references. A declared no-reference operation omits that term. Thus, the conjunction contains the checks selected by the operation's dispatch declaration, and each included check must pass before handler selection. The policy term includes channel tool selection, profile, and duty-separation checks where configured. This condition is necessary for handler selection, not sufficient for actuation: the selected route applies its own guards and approval, then rechecks authority and applicable limits before dispatch.

Fig. 3 gives reference pseudocode for the registered-harness task path, not one literal orchestrator function. Direct REST tool invocation bypasses the task runtime and harness but enters the same shared dispatcher. Route-specific timing, approval, and evidence rules are detailed in Section IV.

## IV. Module Functions and Integration

The architecture in Fig. 1 is implemented as cooperating modules rather than a single monolithic agent. Shared interfaces connect task execution to configuration, industrial

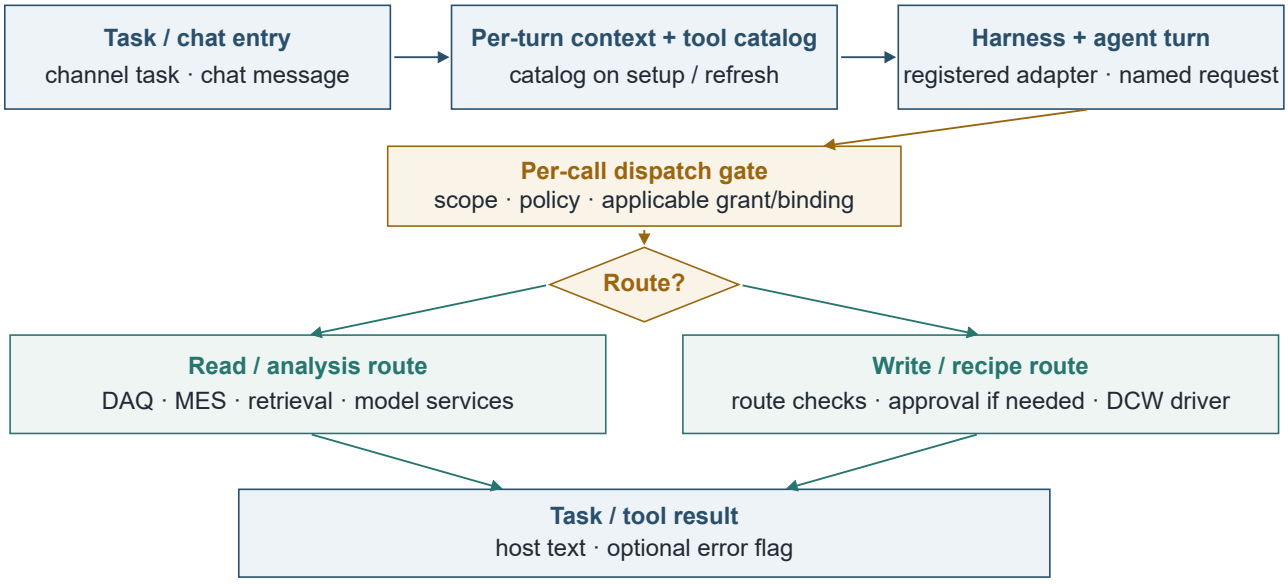



Fig. 2. Registered-harness task execution in AgentWorkShop. The direct REST tool-invocation route is outside this flow and enters the shared dispatcher without a task or harness. The registered harness uses a scoped catalog provisioned at setup and supported registry refreshes; the runtime assembles prompt context per agent turn. The dispatcher checks each returned call for scope, channel policy, and applicable grant and binding conditions before handler selection. Route handlers resolve active-run and approval-dependent state as applicable. The shared tool result contains host text and an optional error flag; route records remain with owning services. Prompt context is not a transactional authorization token. Dispatch-gate errors return before handler execution, whereas route handlers may read state before refusing a write. Durable denial auditing is selective. Receipts and later observations remain distinct.

**Require:** task $q$, channel $c$, bound agent $a$, registered scoped catalog $T$
**Ensure:** task result; any retained evidence remains with its owning service
$x_t \leftarrow \textsc{ResolvePromptContext}(c, q)$
$m \leftarrow \textsc{RetrieveMemory}(q)$
$p \leftarrow \textsc{AssemblePrompt}(q, x_t, m)$
$h \leftarrow \textsc{InvokeRegisteredHarness}(a, p, T)$
**for all** tool request $u$ returned by $h$ **do**
  $g \leftarrow \textsc{CheckScopeAndChannelPolicy}(a, c, u)$
  **if not** $g$ **then**
    $\textsc{EmitToolErrorToHarness}$(h,u); **continue**
  **end if**
  **if** $u$ is an industrial operation **then**
    $v \leftarrow \textsc{CheckGrantBindingOwnership}(a, u)$
    **if not** $v$ **then**
      $\textsc{EmitToolErrorToHarness}$(h,u); **continue**
    **end if**
  **end if**
  $k \leftarrow \textsc{SelectHandler}(u)$
  $r \leftarrow \textsc{ExecuteRouteHandler}(k, a, c, u)$ ▷ handler resolves live state and may persist route-specific records
  $\textsc{EmitDeclaredToolResultToHarness}$(h,r)
**end for**
**return** task state and tool results

Fig. 3. Reference pseudocode for the registered-harness task path. Direct operation invocation is not shown. Evidence persistence is route-specific and remains with the owning service; the shared tool result does not carry a generic evidence envelope.

services, protocol drivers, and route-specific evidence. This section describes those responsibilities and the checks that preserve their boundaries.

The implementation scope is broader than the evaluated scope. Table II separates implemented interfaces from the evidence collected here. An interface intended for extension does not by itself establish interchangeability or portability.

The extension contracts expose the data passed at each boundary. A harness call is normalized as `HostToolCall(toolName, arguments)` and returns `HostToolResult(text, isError)`. A DCW driver accepts an engineering value, tolerance, range, and driver configuration; its result separates write status, raw value, readback, and acknowledgment grade (readback-verified, transport-ack, or unverified). Each DAQ sample stores its node, timestamp, value, and state, with optional line, product, recipe, and run IDs. A write anchor records line/node, prior and requested values, source, actor, time, and optional run/task/approval IDs; recipe definitions retain separate versions. These fields keep tool errors, device receipts, observations, and production lineage inspectable as separate records.

DAQ and write-anchor records may omit line, product, recipe, run, task, or approval identifiers when a route does not supply them. For the R20 trace audit, the join keys are required: task ID, approval ID, recipe-trial run ID, active-run ID, and the run ID returned with the sample must agree with the corresponding route records. R21 checks the corresponding IDs on its deterministic fixture path. Thus the reported complete joins are requirements of those evaluated

records, not a guarantee that every general record contains every identifier.

### A. Task Runtime and Harness Adapters

Task and chat access surfaces create or join a workspace channel, where the task engine manages members, task queues, shared memory, and execution templates. A direct REST tool-invocation endpoint can enter the shared dispatcher without creating a task or invoking a harness; the stdio MCP operation bridge uses this route, while other MCP operations expose management and task services. The harness registry instantiates an adapter behind a common agent interface; the prompt builder combines the task with the active scene brief, industrial context, channel profile, agent binding, and retrieved memory. Tool calls return to the shared dispatcher regardless of which registered harness produced them. The registry is an extension point; the experiments do not establish portability across harnesses. Industrial context is assembled for each agent turn rather than pinned for the task lifetime. Operation handlers check route-specific state at invocation. The implementation does not compare the complete prompt-time line–product–recipe–run snapshot with a task-pinned context version, so the prompt snapshot is grounding information, not a transactional authorization token.

### B. Production Context and Configuration

Scene blueprints describe devices, signal maps, objectives, and process guards. A task resolves these definitions against its active line–product–recipe–run context. The channel profile selects the available capability family, the tool policy narrows the operation catalog, and the agent binding scopes the nodes, controls, and recipe operations available to that member. These filters can restrict one another but cannot broaden the underlying device range. The dispatcher re-reads policy at execution, so a cached catalog does not retain authority after a grant or policy change. The prompt builder assembles the agent's available bound-line and active-run context on each turn; it does not pin the complete context tuple to the task. Specific production snapshot, trial, and MPC routes, together with Hybrid Twin training submissions, resolve and validate a versioned `SceneContract` at operation time. The Twin physics-specification draft route can instead accept a caller-supplied scene description; SceneContract enforcement is therefore route-specific and does not validate or pin all prompt context.

### C. Acquisition, Analysis, and Modeling Services

Semantic DAQ nodes expose typed values, engineering units, ranges, and sampling settings while hiding protocol-specific reads. A per-node DAQ runtime schedules acquisition and stores timestamped observations; the gateway aggregates node pipelines. MES-facing operations provide the configured production-data path. Analysis and retrieval services consume these records, while optional model, training, optimization, and digital-twin services are invoked through their registered operation handlers. These services complement the agent workflow; AgentWorkShop does not prescribe a single optimization strategy.

### D. Semantic I/O and Actuation

The driver registry resolves a semantic node to a built-in or plugin driver. DAQ handlers use the node's acquisition interface; DCW handlers route a write or readback through the controller and the configured protocol driver. This separation lets recipes and agents refer to semantic controls without embedding Modbus, OPC UA, MQTT, HTTP, or MES transport details in their operation arguments. The supported driver mappings are implementation interfaces; only the mappings and workflows enumerated in Section V are experimentally exercised.

### E. Shared Dispatch and Write Admission

Every named operation passes the dispatcher. Its ordered checks validate invocation scope and current channel policy (tool selection, profile, and applicable duty separation), then the agent's binding and runtime grant and ownership of referenced resources. A refusal returns the excluding condition. Denial persistence is selective: grant denials and selected resource-reference denials have explicit audit paths, whereas scope and channel-policy denials are not uniformly retained as durable audit events. A catalog entry therefore describes availability but does not authorize execution. Handlers for operations that require manual approval suspend for a decision, then revalidate authority and active limits before dispatch. For manual recipe operations, the handler checks the active run, candidate parameters, and route timing before approval; after the wait it rechecks current authority and active limits. For `recipe_apply`, a changed definition requires a new proposal. These route-specific checks complement, but do not replace, the per-call dispatcher checks.

All actuator writes converge on the DCW admission point. For an online node write, the admissible interval is

$$\mathcal{I}_n^{\text{online}}(t) = \mathcal{H}_n \cap \mathcal{B}_n \cap \mathcal{P}_n(t) \cap \mathcal{R}_n(t), \tag{2}$$

where $\mathcal{H}_n$ is the hard engineering range and $\mathcal{B}_n$, $\mathcal{P}_n$, and $\mathcal{R}_n$ are the baseline, active-product, and active-recipe limits. Missing optional limits impose no restriction, whereas an empty intersection rejects the request and identifies the excluding layer. Recipe dispatch applies the hard, baseline, and product limits to the replacement recipe rather than intersecting with the previous recipe window. The actuator path also enforces step, interval, and hold-window checks. These admission rules constrain commands; process objectives and coupled variable guards are assessed from later observations. The framework does not replace independent PLC or safety-instrumented protection.

### F. Integrated Operation and Route-Specific Evidence

A recipe trial illustrates how the modules cooperate. The task runtime delivers a named `recipe_trial` call from the agent harness; the dispatcher checks the agent's recipe binding and execution scope; the recipe handler checks the active run, candidate parameters, and route timing; and a manual binding requests approval before the DCW controller sends the batch through its driver. The handler returns a per-parameter receipt.

TABLE II
IMPLEMENTED INTERFACES AND EVIDENCE COVERAGE. "NOT EVALUATED" MARKS FEATURES OUTSIDE THE REPORTED EXPERIMENTS; IT DOES NOT IMPLY THAT THEY ARE ABSENT FROM THE IMPLEMENTATION.

| Path | Implemented interface and checks | Evidence in this study |
|---|---|---|
| Task and harness | Registered harness adapters and task/chat services; direct REST operation calls share the dispatcher but bypass the task runtime. | R20 exercises six registered-harness tasks; R21 exercises one deterministic fixture task per scene. Alternate harnesses and direct REST calls are not evaluated. |
| Dispatch and write | Every call checks scope, policy, grant, binding, and resource ownership as applicable. Recipe trials check active run, candidate bounds, and timing before approval, then recheck authority and active limits; changed recipe definitions require a new apply proposal. | R20/R21 exercise approved recipe trials and verified readback. Mock tests cover denied/expired approval and changed recipe/limit state. Online node writes are not evaluated end to end. |
| Semantic I/O | DAQ/DCW nodes resolve configured semantic signals to built-in or plugin drivers. | R20/R21 exercise two scene mappings; R13 provides exploratory protocol probes. The normal online DAQ backend and physical equipment are not evaluated. |
| Records | Owning services retain task, approval, receipt, observation, and recipe-version records separately; production identifiers are route-dependent. | R20 reports six complete task-to-sample joins; R21 reports two fixture-path joins. Neither result establishes universal record completeness across untested routes. |

The caller then obtains DAQ observations and can request a separate update, apply, or rollback operation. A trial receipt confirms command handling; it does not by itself establish process attainment or persist a new recipe version.

The routes retain different records because they answer different questions. A driver acknowledgment reports command evidence; DAQ reports a process observation; an exploration summary records a bounded probe; and a recipe version records a saved definition. Their relation through actor and production context allows an execution to be inspected without treating these outcomes as interchangeable.

### G. Route-Specific Lifecycles and Evidence

*a) Timing and approval:* Timing guards are route-specific. Online node writes enforce an effective minimum spacing of 60 s. Recipe dispatch uses a per-recipe interval anchored to approval, while trials also use a configurable per-line cadence (default 300 s). The recipe interval can refuse a proposal before an approval request is created. Checks occur at proposal or execution according to the operation, and experiment-specific pacing overrides are reported with the corresponding results.

Manual bindings create an approval request containing the actor, node or recipe, candidate values, applicable limits, expiry, and parameter bases. The default wait expires after 180 s and refuses execution. After approval, the executor rechecks authorization and active limits to account for context changes during the wait. For `recipe_apply`, a changed definition requires a new proposal; an unchanged approval dispatches a copy of the approved parameter snapshot. A line-level recipe mode defaults to manual; changing it to automatic requires administrator confirmation. The approval record authorizes a workflow action and does not replace process protection.

*b) Node probe:* The bounded `optimization_explore` route is exposed to the AML optimization profile. It validates a bound control and DAQ scope, captures a baseline, routes a bounded write through DCW, and stores a later target window as a separate exploration summary. Its timing and approval behavior is route-specific; R20 does not exercise this route.

*c) Recipe operations:* `recipe_trial` checks the bound recipe and active run, then dispatches candidate overrides as a batch with per-parameter verification. The caller subsequently queries DAQ and may save a definition with `recipe_update`, restore one with `recipe_rollback`, or apply a version with `recipe_apply`. A trial does not save or apply a version, and endpoint verification is not a process-response measurement. Public direct-node control is closed; bounded probes and recipe operations are the configured write surfaces.

*d) Command verification:* Command confirmation and process response answer different questions. A driver ACK is graded as `readback-verified`, `transport-ack`, or `unverified`; a matching endpoint read supports the first grade, whereas transport or MES application confirmation supports the second. Recipe-batch verification is `verified`, `unverified`, or `failed`. If readback is absent or mismatched, up to three reads are attempted 2 s apart. Initial ACKs, final ACKs, and per-parameter verdicts are retained separately. A readback-verified setpoint does not establish target attainment.

*e) Observation, verdict, and compensation:* Recipe-path observation follows batch dispatch, and saving a definition, applying it, and observing process response remain separate operations. An open node record can request rollback after repeated breaches; missing samples do not count as breaches. Caller verdicts and compensation remain separate from the observation and subject to admission checks.

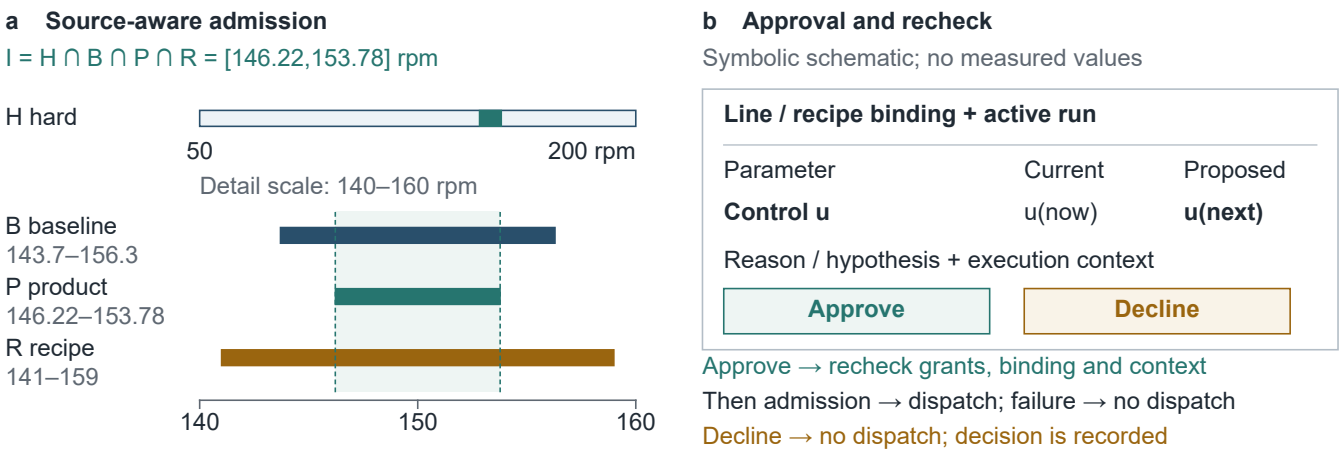



Fig. 4. Admission and approval boundaries. (a) Illustrative archived RPM configuration (not part of R20): online node interventions use the four-layer value interval of Eq. (2), whereas recipe dispatch omits the prior recipe window. (b) Symbolic human approval binds a proposed action and authority fields; execution checks run again after approval, which does not reserve equipment state.

## V. Experimental Evaluation

The evaluation asks whether one registered-harness path can execute bounded tasks across two simulator scenes with different signal and protocol mappings while retaining a checkable task-to-approval-to-receipt-to-sample trace. R20 is the primary result: six sequential tasks through one registered harness, three per scene on one simulator instance per scene. It demonstrates this configured software path, not configuration-reuse benefit or process attainment. R21 is a separate deterministic fixture replay on an isolated current-version application and simulator; it checks the same task-to-device path without an LLM decision. R13 is secondary exploratory protocol and data-path evidence whose exact runtime source was not pinned. All runs use repository simulators. Supplementary File S1 contains the R20 trace matrix and records, plus the R21 fixture evidence.

### A. Evidence Units and Experimental Procedure

R20 uses the registered `omp` harness in cast-film and annealing scenes, with three sequential task executions per scene on one simulator instance per scene. R13 uses configured protocol and data paths in the same two simulator scenes. Earlier experiments are retained separately in Supplementary File S1 and are not pooled with these results.

### B. Registered-Harness Process Path (R20)

R20 ran the local v0.8.0 working-tree application with the registered `omp` adapter and GLM-5.3-Flash. Supplementary File S1 contains the recorded base revisions, working-tree patches, dependency lockfiles, prompt assets, and run logs. This is a source-reconstruction kit rather than an immutable release, and the external model service does not guarantee deterministic replay. Each of six sequential executions created a separate line, product, recipe, run, channel, agent, and task context. Three executions shared one simulator instance per scene. Cast film used OPC UA for screw-speed control and MQTT for thickness; annealing used Modbus TCP for both the zone-3 setpoint and strip-temperature probe. The model received a live control read, process measurement, and target band, then selected one value from the scenario-defined prevalidated set (125/130/135 rpm or 735/740/745 °C) for a bounded `recipe_trial`; this was a constrained choice, not an unrestricted optimization search. A synthetic administrator approved each proposal. Each route log contains one trial and a 1/1 verified device receipt.

After verifying the route ACK and returned trial result, the runner initiated a 20-s delay, then probed, ingested, and queried one value per task under the active-run tag through the isolated SQLite time-series fallback. The runner checked task state after the sample query; all six task records were then `COMPLETED`. The runner reported 76 passing assertions and no failures, and the separate trace audit found a complete task-to-sample record in all six executions: each had a matching approval ID, one readback-verified trial receipt, and one query result whose run ID and value matched the active run and explicit probe. The task-completion timestamp was not retained, so no task-completion-to-sample interval can be computed. Table III labels Film-1–3 map to `cast-film-r1--r3`, and Anneal-1–3 to `annealing-r1--r3`; full identifiers and per-record checks are listed in `R20-TRACE-MATRIX.csv` in S1.

The trace matrix distinguishes two run identifiers. The trial run ID denotes the `recipe_trial` operation record; the active run ID denotes the production run used to tag and query the probe. Both are linked to the same task and approval record, and they are not expected to be equal.

Table III reports the pre-task process value, selected setting, and run-tagged sample for each execution. All six samples remained outside target. The experiment verifies the bounded harness-to-simulator execution and record path, not process attainment or convergence. Task completion denotes a completed software workflow, not satisfaction of the manufacturing objective. A single probe is not a response trajectory, and the SQLite fallback does not test the normal online DAQ backend. R20 also does not isolate profile/binding reuse or establish cross-harness portability.

TABLE III
CONSTRAINED MODEL SELECTIONS AND SIMULATOR MEASUREMENTS. THE CONTROL IS SCREW SPEED FOR CAST FILM AND THE ZONE-3 FURNACE SETPOINT FOR ANNEALING. EACH ROW REPORTS ONE RUN-TAGGED PROBE AFTER A 20-S DELAY FROM VERIFIED ROUTE READBACK. ALL SIX SAMPLES ARE OUTSIDE TARGET; EXECUTIONS ARE SEQUENTIAL WITHIN EACH SCENE.

| Case | Pre-task value / target band | Control: start→trial | Sample |
|---|---|---|---|
| Film-1 | 55.82/48–52 $\mu$m | 120 → 125 rpm | 45.69 $\mu$m |
| Film-2 | 45.69/48–52 $\mu$m | 120 → 130 rpm | 46.69 $\mu$m |
| Film-3 | 46.69/48–52 $\mu$m | 120 → 125 rpm | 45.69 $\mu$m |
| Anneal-1 | 675.40/700–760 °C | 730 → 745 °C | 679.60 °C |
| Anneal-2 | 678.30/700–760 °C | 730 → 745 °C | 677.70 °C |
| Anneal-3 | 676.90/700–760 °C | 730 → 745 °C | 678.10 °C |

### C. Deterministic Fixture Replay (R21)

R21 ran the local v0.8.0 application with the registered OMP transport fixture, which returned a fixed tool call without invoking an LLM or autonomous harness decision. In one fresh isolated application/simulator environment, one fixture task per scene invoked the recipe-trial path in cast-film and annealing. The runner checked dispatch, synthetic approval, verified receipt, matching device readback, run-tagged DAQ probe, and route log; all 31 runner assertions, including process and port cleanup checks, passed. This replay shows that the configured software route can be exercised with deterministic harness input. It is not an independent real-agent run, a process replication, or evidence of target attainment. The two fixture tasks are not pooled with R20.

### D. Software Contract Checks

A separate regression run on the local v0.8.0 working tree records 65 passing software tests across eight suites and no failures. Denied or expired approvals, changed recipe or limit state after approval, and invalid node values are refused before driver actuation; unchanged approved state dispatches through the mock driver and readback. The suites also cover tool scope, grants, and role-specific exposure. These checks use temporary or mock state and are separate from R20's 76 runner checks; they do not represent process trials or test missing/mismatched DAQ run tags in the end-to-end path.

### E. Exploratory Protocol and Data Path (R13)

The R13 application report describes five protocol-family read probes (OPC UA, MQTT, Modbus TCP, Modbus RTU over TCP, and HTTP/REST) and bounded writes with matching readbacks in two simulator scenes. Six sequential 120-s windows per scene at 1-s sampling produced lineage-verified datasets with 658 cast-film rows and 655 annealing rows. The rows are overlapping time-window samples, not independent process replications. The exact runtime source was not recorded. R13 is used only as exploratory protocol and data-lineage evidence; its archived gate outputs and run-boundary audit are documented in the reproducibility supplement and do not support the primary registered-harness result. R14 is an earlier exploratory pilot with alternative signal pairs; both pilot model gates failed, and those outcomes informed the subsequent R13 pair selection. R13 also produced no eligible model, recommendation, or dispatch. The two runs remain separate and are not model-performance evidence.

Earlier protocol, admission-ablation, controller, deployment, and scripted recipe records in S1 use older versions or interfaces. They are not pooled with R13 or R20, used as matched strategy comparisons, or relied on for the current integration claims.

## VI. DISCUSSION

R20 demonstrates the bounded registered-harness path across the two tested simulator scenes: tasks completed with synthetic approval and verified readback, and all six records passed the task-to-approval-to-receipt-to-sample link checks. R21 exercised one deterministic fixture task per scene and passed its 31 runner assertions, including the run-tagged sample checks. No R20 post-trial sample reached its target band. The 20-s sample delay began after receipt/readback verification; the runner checked task state after the sample query, and no task-completion-to-sample interval is retained. R13 separately contributes exploratory protocol and data-lineage evidence. Together, these runs support only the configured software paths; they do not establish process-control efficacy or portability.

R20 uses one simulator instance per scene and does not establish independent replication, convergence, or quality improvement. Its single post-trial sample per execution is not a response trajectory or a task-completion-to-sample latency measurement. Controlled profile/binding adaptation, cross-harness portability, normal online DAQ backend behavior, and physical transfer remain untested. The synthetic approval records do not measure operator effectiveness [31]–[33]; restricted tool access does not establish deployment security [26], [34], [35] or replace independently engineered PLC/SIS protection [36].

Future evaluation should use independent simulator replications, a controlled configuration-change test, the normal online DAQ backend, and physical-plant trials. These tests are needed to assess benefits beyond the configured paths reported here.

## VII. CONCLUSION

AgentWorkShop provides a modular reference implementation that connects collaborative tasks, production context, operation checks, semantic I/O, and route-specific records. R20 demonstrates a live-agent task-to-probe-and-ingest trace in two simulator scenes, with complete linkage checks for six executions; each post-readback sample was a single snapshot through the isolated SQLite fallback, and none reached target. R21 separately verifies the deterministic task route in both scenes, while R13 supplies exploratory protocol and data-path evidence. These results establish behavior of the tested software paths, not process-control efficacy, comparative advantage, or portability.

## REFERENCES

[1] Y. Xia, N. Jazdi, J. Zhang, C. Shah, and M. Weyrich, "Control industrial automation system with large language model agents," in *Proc. IEEE 30th Int. Conf. on Emerging Technologies and Factory Automation (ETFA)*, 2025, pp. 1–8, doi: 10.1109/ETFA65518.2025.11205539.

[2] K.-C. Huang, H.-T. Lin, C.-H. Lin, and M.-T. Huang, "Large language model integration in industrial cyber-physical systems: A multi-modal architecture for automated control generation," in *Proc. Int. Conf. on Control, Automation and Diagnosis (ICCAD)*, 2026, pp. 1–6, doi: 10.1109/ICCAD69956.2026.11643124.

[3] K. Köhle, D. Anicic, T. A. Runkler, and R. Graf, "LLM-driven multi-agent control for skill-based smart manufacturing," in *Proc. IEEE 31st Int. Conf. on Emerging Technologies and Factory Automation (ETFA)*, 2026, pp. 1–8, doi: 10.1109/ETFA68377.2026.11725257.

[4] M. Waseem, K. Bhatta, C. Li, and Q. Chang, "Pretrained LLMs as real-time controllers for robot operated serial production line," *Mach. Learn.: Eng.*, vol. 2, no. 1, 2026, art. no. 015006; doi: 10.1088/3049-4761/ae4ee8.

[5] Z. Liu *et al.*, "Agents4PLC: Automating closed-loop PLC code generation and verification in industrial control systems using LLM-based agents," *IEEE Trans. Softw. Eng.*, vol. 52, no. 5, pp. 1672–1687, 2026, doi: 10.1109/TSE.2026.3667895.

[6] R. Wu, J. Ai, and T. Li, "InstructMPC: A human-LLM-in-the-loop framework for context-aware control," in *Proc. 2025 IEEE 64th Conference on Decision and Control (CDC)*, 2025, pp. 172–179, doi: 10.1109/CDC57313.2025.11312156.

[7] J. Lim and I. Kovalenko, "A large language model-enabled control architecture for dynamic resource capability exploration in multi-agent manufacturing systems," in *Proc. IEEE 21st Int. Conf. on Automation Science and Engineering (CASE)*, 2025, pp. 2088–2095, doi: 10.1109/CASE58245.2025.11163802.

[8] Z. Zhao *et al.*, "A large language model-based multi-agent manufacturing system for intelligent shopfloors," *Adv. Eng. Inform.*, vol. 69, 2026, art. no. 103888; doi: 10.1016/j.aei.2025.103888.

[9] M. S. Gill, J. Vyas, A. Markaj, F. Gehlhoff, and M. Mercangöz, "Leveraging LLM agents and digital twins for fault handling in process plants," in *Proc. IEEE ETFA*, 2025, pp. 1–8, doi: 10.1109/ETFA65518.2025.11205597.

[10] B. Hofmann, N. Piechulek, S. Kreitlein, J. Franke, and P. Bruendl, "Beyond touch-based human-machine interface: Control your machines in natural language by utilizing large language models and OPC UA," *arXiv preprint arXiv:2510.11300*, 2025. [Online]. Available: https://arxiv.org/abs/2510.11300v2

[11] C. Graul, W. Żarski, D. Mikołajewski, and I. Rojek, "Ai agents in industry 4.0: AAS–OPC UA–LLM architecture as the foundation of intelligent manufacturing systems in the context of industrial enterprise implementation," *Applied Sciences*, vol. 16, no. 11, p. 5428, 2026, doi: 10.3390/app16115428.

[12] D. Schall, "Safe integration of large language models into industrial process control: A multi-agent architecture with P&ID-grounded validation," *Autonomous Intelligent Systems*, vol. 6, 2026, article no. 14; doi: 10.1007/s43684-026-00136-1.

[13] S.-C. Lin and C.-Y. Chen, "An integrated framework featuring policy-governed agentic AI for closed-loop manufacturing control with multi-source sensor–MES–ERP," *The International Journal of Advanced Manufacturing Technology*, vol. 143, pp. 6405–6428, 2026, doi: 10.1007/s00170-026-17806-2.

[14] A. Biskupovic, A. Villalonga, F. Castaño, R. E. Haber, and F. Núñez, "Agent-based supervision for service-oriented industrial cyber–physical systems," *IEEE Trans. Ind. Inform.*, vol. 21, no. 3, pp. 2719–2728, 2025, doi: 10.1109/TII.2024.3514178.

[15] Y. Xia, N. Jazdi, and M. Weyrich, "An architecture for integrating large language models with digital twins and automation systems," in *2025 IEEE 30th International Conference on Emerging Technologies and Factory Automation (ETFA)*, 2025, pp. 1–8, doi: 10.1109/ETFA65518.2025.11205636.

[16] L. M. Vieira da Silva, A. Köcher, and F. Gehlhoff, "Beyond formal semantics for capabilities and skills: Model context protocol in manufacturing," in *2025 IEEE 30th International Conference on Emerging Technologies and Factory Automation (ETFA)*, 2025, pp. 1–4, doi: 10.1109/ETFA65518.2025.11205601.

[17] K. Köhle, D. Anicic, T. A. Runkler, K. Dorofeev, A. S. Thuluva, and R. Graf, "Virtually commissioned semantic recipes based on OPC UA skills," in *2025 IEEE 8th International Conference on Industrial Cyber-Physical Systems (ICPS)*, 2025, pp. 1–6, doi: 10.1109/ICPS65515.2025.11087832.

[18] X. Jiang *et al.*, "Agentic AI for safety-aware process monitoring and fault diagnosis: A review," *Processes*, vol. 14, no. 13, 2026, art. no. 2112; doi: 10.3390/pr14132112.

[19] Model Context Protocol contributors, "Model Context Protocol: Tools," Specification, version 2026-07-28, 2026, accessed: Oct. 20, 2026. [Online]. Available: https://modelcontextprotocol.io/specification/2026-07-28/server/tools

[20] ISA, "ISA-88.00.01-2010: Batch control—part 1: Models and terminology," International Standard, 2010.

[21] ——, "ANSI/ISA-95.00.01-2025 (IEC 62264-1 mod): Enterprise-control system integration—part 1: Models and terminology," International Standard, 2025.

[22] J. Moyne, E. del Castillo, and A. M. Hurwitz, Eds., *Run-to-Run Control in Semiconductor Manufacturing*, 1st ed. Boca Raton, FL: CRC Press, 2000.

[23] H. Wang, C. M. Poskitt, and J. Sun, "AgentSpec: Customizable runtime enforcement for safe and reliable LLM agents," in *Proc. IEEE/ACM 48th Int. Conf. on Software Engineering (ICSE)*, 2026, pp. 2938–2950, doi: 10.1145/3744916.3764546.

[24] E. Debenedetti *et al.*, "Defeating prompt injections by design," 2025, arXiv preprint arXiv:2503.18813. [Online]. Available: https://arxiv.org/abs/2503.18813

[25] U.S. Food and Drug Administration, "Part 11, electronic records; electronic signatures—scope and application," Guidance for Industry, 2003.

[26] IEC, "IEC 62443-3-3:2013: Industrial communication networks—network and system security—part 3-3: System security requirements and security levels," Standard, 2013.

[27] IEEE, "IEEE std 1815-2012: Standard for electric power systems communications—distributed network protocol (DNP3)," Standard, 2012.

[28] M. Alshiekh, R. Bloem, R. Ehlers, B. Könighofer, S. Niekum, and U. Topcu, "Safe reinforcement learning via shielding," in *Proc. AAAI Conference on Artificial Intelligence*, vol. 32, no. 1, 2018, pp. 2669–2678, doi: 10.1609/aaai.v32i1.11797.

[29] A. D. Ames, X. Xu, J. W. Grizzle, and P. Tabuada, "Control barrier function based quadratic programs for safety critical systems," *IEEE Trans. Autom. Control*, vol. 62, no. 8, pp. 3861–3876, 2017, doi: 10.1109/TAC.2016.2638961.

[30] L. Sha, "Using simplicity to control complexity," *IEEE Softw.*, vol. 18, no. 4, pp. 20–28, 2001, doi: 10.1109/MS.2001.936213.

[31] R. Parasuraman and D. H. Manzey, "Complacency and bias in human use of automation: An attentional integration," *Hum. Factors*, vol. 52, no. 3, pp. 381–410, 2010, doi: 10.1177/0018720810376055.

[32] R. Parasuraman, T. B. Sheridan, and C. D. Wickens, "A model for types and levels of human interaction with automation," *IEEE Trans. Syst., Man, Cybern., Part A*, vol. 30, no. 3, pp. 286–297, 2000, doi: 10.1109/3468.844354.

[33] L. Bainbridge, "Ironies of automation," *Automatica*, vol. 19, no. 6, pp. 775–779, 1983, doi: 10.1016/0005-1098(83)90046-8.

[34] K. Greshake, S. Abdelnabi, S. Mishra, C. Endres, T. Holz, and M. Fritz, "Not what you've signed up for: Compromising real-world LLM-integrated applications with indirect prompt injection," in *Proc. 16th ACM Workshop on Artificial Intelligence and Security*, 2023, pp. 79–90, doi: 10.1145/3605764.3623985.

[35] Y. Liu, Y. Jia, R. Geng, J. Jia, and N. Z. Gong, "Formalizing and benchmarking prompt injection attacks and defenses," in *Proc. 33rd USENIX Security Symposium (USENIX Security 24)*, 2024, pp. 1831–1847. [Online]. Available: https://www.usenix.org/conference/usenixsecurity24/presentation/liu-yupei

[36] IEC, "IEC 61511-1:2016: Functional safety—safety instrumented systems for the process industry sector—part 1: Framework, definitions, system, hardware and application programming requirements," International Standard, 2016.